



1. **What makes up the upper airway?:** The larynx and up
2. **What makes up the lower airway?:** The pharynx and below
3. **What is surfactant?:** a substance in the alveoli that reduces surface tension, makes gas exchange easier, and prevents atelectasis.
4. **What is our primary respiratory center?:** Medulla Oblongata
5. **Where are chemoreceptors, and what do they detect?:** Primary chemoreceptors in the medulla oblongata, primarily triggered by H^+
6. **What is diffusion?:** Movement of a solute from a high to a low concentration.
7. **Where is the phrenic nerve?:** C3-C5
8. **Describe Inhalation:** An active process where the diaphragm contracts expanding the chest cavity. Negative pressure will force the air to rush in.
9. **What is the primary trigger for respiration?:** Increased CO_2 , Increased H^+ , decreased Ph .
10. **Describe exhalation:** A passive process where the muscles relax and recoil air out of the lungs. Positive pressure forces air out of the chest cavity.
11. **What core muscles are activated during inspiration?:** Diaphragm, external intercostals
12. **What core muscles are activated during expiration?:** Diaphragm, internal intercostals
13. **What accessory muscles are activated during inspiration?:** Sternocleidomastoid, Pectoralis Major
14. **What accessory muscles are activated during expiration?:** Abdominal muscles, quadratus lumborum
15. **What is paradoxical breathing?:** abnormal breathing where the chest and abdomen move in opposite directions. When the diaphragm is weakened, the abdomen will move inward during inspiration, and outward during expiration
16. **Ventilation is mainly affected by...:** Tidal volume and rate
17. **Oxygenation is mainly affected by...:** FiO_2 and PEEP
18. **What is tidal volume?:** Amount of air in each breath
19. **What is minute volume?:** Tidal volume $\times$ RR. the total amount of air inhaled or exhaled from a person's lungs in one minute
20. **Dead Space:** volume of inhaled air that does not participate in gas exchange
21. **Residual Volume:** Remaining air after max exhalation
22. **V/Q mismatch:** A mismatch between Ventilation and Perfusion. (Adequate ventilation, poor perfusion) (PE)
23. **What is the best indicator of declining respiratory status?:** Mental Status



24. **What does rhonchi often represent:** thick obstructions/secretions
25. **What do rales/crackles often represent:** Fluid/secretions in airway
26. **What 3 things make up asthma?:** Bronchospasm, Bronchoedema, and Bronchorrhea.
Smooth muscle constriction, inflammation, and hypersecretion of mucus via goblet cells.
27. **What can we use to treat asthma?:** O₂, CPAP, Bronchodilation, Ipratropium, and nebulized epi.
28. **How long does albuterol last compared to epi when nebulized?:** Albuterol can last for 3-6 hours when nebulized, while epi can last about 30 minutes.
29. **What is our starting PEEP for Asthma and CPAP?:** Starting PEEP at 5
30. **Can we use a nebulizer in CPAP?:** Yes
31. **What causes COPD?:** Irritation/Inflammation due to chronic exposure to irritants (smoking #1)
32. **How is alveoli affected by COPD?:** Alveoli face irreversible destruction of Alveolar walls and capillary beds, leading to a loss of surface area for gas exchange, and a loss of elastic recoil.
33. **Someone presenting with pursed lip breathing, pink skin, and diminished lung sounds might be suffering from...:** Emphysema
34. **How are bronchioles affected by COPD?:** Chronic inflammation leads to a remodeling of the Bronchioles, causing fibrosis, Smooth muscle hypertrophy, and goblet cell hypertrophy
35. **A patient presenting with a productive cough, cyanotic appearance, and edema in legs might be suffering from:** Chronic Bronchitis
36. **How do we treat COPD?:** O₂, CPAP splinting, Bronchodilation, Anticholinergic, position of comfort.
37. **CPAP _____ Intrathoracic Pressure:** Increases
38. **How can a patient benefit from PEEP?:** PEEP increases surface area for gas exchange, leaving the lungs partially inflated and re expanding atelectasis.
39. **What are contraindications of CPAP?:** Hypoperfusion, Pneumothorax, Facial trauma, AMS/LOC, Respiratory arrest
40. **A patient presenting with Vomiting, hypotension, and severe SOB/Wheezing might be suffering from...:** Anaphylaxis
41. **What is anaphylaxis?:** A life threatening allergic reaction affecting 2 or more body systems.
42. **What are our 3 main Inflammatory Mediators?:** Histamine, Leukotrienes, Cytokines
43. **How can we treat anaphylaxis:** Epinephrine, Benadryl, Albuterol.
44. **What are acute causes of CHF?:** Myocardial Ischemia, Myocarditis
45. **What are chronic causes of CHF:** Hypertension, Valvular Disease



46. **If we have left sided heart failure, where will our blood back up?:** The left atrium, and then eventually the pulmonary system
47. **If we have right sided heart failure, where will our blood back up?:** Right Atrium, Vena Cava, and eventually our venous system
48. **What 3 things can most commonly exacerbate CHF?:** Diet Changes 48%
Non Compliance with medication 28%
New prescription 15%
49. **What is the effect of Increasing Sodium levels?:** An increase in circulating volume
50. **What is an effect of increased circulating volume?:** Increased Preload
51. **How does CHF and RAAS relate?:** The kidney will release renin to improve SVR, increasing circulating volume, backing fluid into the pulmonary system, and causing pulmonary hypertension fluid shifts into the alveoli
52. **What is treatment for CHF?:** Nitroglycerin, CPAP
53. **What is NTG mechanism of action?:** NTG is converted into nitric oxide. Nitric oxide acts as a potent vasodilator.
54. **How is angiotensin II different from NTG?:** Angiotensin II causes vasoconstriction by decreasing Nitric Oxide production
55. **A patient presenting with Fever, Isolated Rhonchi, and SOB may be suffering from?:** Pneumonia
56. **A tripodizing child presenting with fever, drooling, and a hoarse voice may be suffering from:** Epiglottitis
57. **How can we help treat Epiglottitis?:** Blow by O2 with Saline Nebulizer, early ALS, calming the patient.
58. **A patient presenting with Stridor on inspiration, and a seal like bark may be suffering from:** Croup
59. **How can we treat croup?:** Nebulized Saline or Epi, O2, calming the patient
60. **Why is RSV life threatening for Infants?:** Causes Apnea and Lethargy
61. **What is virchows triad?:** Venous Stasis
Endothelial Injury
Hypercoagulability
62. **What are signs of Hyperventilation?:** Numbness around lips, carpopedal spasms, dizziness, chest pain.